Transformers-ACE

A Strictly Local Equivariant ACE-Attention Interatomic Potential

Methods and Architecture Specification, Version 2

Transformers-ACE project

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Abstract

Transformers-ACE is a local, energy-conserving machine-learned interatomic potential that combines compressed atomic-cluster-expansion (ACE) density correlations with equivariant neighbor-set attention. The architecture is built for scalable molecular dynamics without atom-to-atom propagation of updated hidden states. Periodic geometry enters through shifted displacement vectors, chemical and angular edge densities are represented in irreducible representations of $O(3)$, and learned Clebsch–Gordan products produce correlations through a configurable maximum body order. Attention queries depend on the current center, whereas keys and equivariant values depend only on fixed local edge features. A compact C^2 cutoff and a null channel in the attention normalization make energy and forces continuous when an edge leaves the neighbor list. A scalar total energy is the only predicted observable; forces and stress are exact derivatives of that energy. This document specifies the equations implemented by architecture version 2, the training objective, locality and equivariance properties, checkpoint semantics, and validation requirements.

1 Design scope

The objective is an accurate short-range potential with linear system-size scaling and conservative dynamics. The implementation follows four constraints:

1. all geometric interactions are restricted to a finite cutoff;
2. every intermediate feature has a defined $O(3)$ transformation law;
3. no updated hidden state of atom j is transmitted to atom i;
4. energy is a scalar and all mechanical observables are its derivatives.

The strictly local construction is motivated by local equivariant potentials such as Allegro [3], while the representation algebra uses the ACE and Euclidean-neural-network frameworks [1, 2]. Transformers-ACE is not an EquiformerV2 implementation: it retains compact $\ell_{\max} = 2$ channels and a local edge-attention mechanism rather than a deep high-degree message-passing transformer [4].

2 Periodic local environments

Let atom i have Cartesian row vector $\boldsymbol{r}_i$, atomic number Z_i, and a periodic cell matrix $\boldsymbol{h}$. ASE supplies an integer image shift $\boldsymbol{S}_{ij} \in \mathbb{Z}^3$ for every directed edge $j \to i$. The periodic displacement and distance are

$$\boldsymbol{r}_{ij} = \boldsymbol{r}_j - \boldsymbol{r}_i + \boldsymbol{S}_{ij}\boldsymbol{h}, \tag{1}$$

$$d_{ij} = \|\boldsymbol{r}_{ij}\|_2, \qquad \widehat{\boldsymbol{r}}_{ij} = \boldsymbol{r}_{ij}/d_{ij}. \tag{2}$$

The neighbor set is

$$\mathcal{N}(i) = \{j, \boldsymbol{S}_{ij} : d_{ij} < r_{\mathrm{c}}\}. \tag{3}$$

The directed convention is important: spherical harmonics and forces are evaluated from neighbor to center using the same shifted vector in training and ASE inference.

3 Smooth radial basis

Architecture v2 uses the compact quintic envelope

$$f_{\mathrm{c}}(d) = \begin{cases} 1 - 10x^3 + 15x^4 - 6x^5, & 0 \le x < 1, \\ 0, & x \ge 1, \end{cases} \qquad x = d/r_{\mathrm{c}}. \tag{4}$$

It satisfies

$$f_{\mathrm{c}}(r_{\mathrm{c}}) = f_{\mathrm{c}}'(r_{\mathrm{c}}) = f_{\mathrm{c}}''(r_{\mathrm{c}}) = 0, \tag{5}$$

which is required for smooth forces, stress, phonons, and cell dynamics at the neighbor-list boundary. The default radial basis is

$$B_n(d) = f_{\mathrm{c}}(d)\sqrt{\frac{2}{r_{\mathrm{c}}}}\frac{\sin(n\pi d/r_{\mathrm{c}})}{d}, \qquad n = 1, \dots, N_r, \tag{6}$$

with the analytic $d \to 0$ limit. A Gaussian basis multiplied by the same envelope is also available. For the $CsPbI_3$ configuration, $r_{\mathrm{c}} = 6$ Å and $N_r = 12$.

4 Equivariant representations

A feature transforming in irrep (ℓ, p) is denoted $x_m^{(\ell,p)}$, where $m = -\ell, \dots, \ell$ and $p \in \{+1, -1\}$ is parity. Under $g \in O(3)$,

$$x^{(\ell,p)} \mapsto D^{(\ell,p)}(g)x^{(\ell,p)}. \tag{7}$$

Spherical harmonics $Y_{\ell m}(\widehat{\boldsymbol{r}}_{ij})$ carry parity $p = (-1)^\ell$. Clebsch–Gordan tensor products combine two irreps according to the triangle and parity rules,

$$[x \otimes_{\mathrm{CG}} y]_m^{(\ell,p)} = \sum_{\ell_1 m_1, \ell_2 m_2} W_{c_1 c_2 c}^{\ell_1 \ell_2 \ell} C_{\ell_1 m_1, \ell_2 m_2}^{\ell m} x_{c_1 m_1}^{(\ell_1, p_1)} y_{c_2 m_2}^{(\ell_2, p_2)}, \quad p = p_1 p_2. \tag{8}$$

The coefficients C are fixed by symmetry and the multiplicity mixing weights W are learned. The implementation uses e3nn tensor products [2].

5 Chemical-angular edge density

Atomic numbers are mapped to scalar embeddings $\boldsymbol{e}_i = \mathrm{Emb}(Z_i) \in \mathbb{R}^{C_0}$. A radial network maps $\boldsymbol{B}(d_{ij})$ to edge-dependent tensor-product weights. The edge density is

$$\boldsymbol{a}_{ij} = \mathrm{TP}_\rho\Big(\boldsymbol{e}_j, \{Y_{\ell m}(\widehat{\boldsymbol{r}}_{ij})\}_{\ell=0}^{\ell_{\max}}; \mathrm{MLP}_r[\boldsymbol{B}(d_{ij})]\Big). \tag{9}$$

Because Eq. (6) includes f_{c}, every component of $\boldsymbol{a}_{ij}$ vanishes smoothly at the cutoff. The permutation-invariant local density at center i is

$$\boldsymbol{A}_i = \sum_{j \in \mathcal{N}(i)} \boldsymbol{a}_{ij}. \tag{10}$$

For the default v2 model, the compact correlation irreps are

$$16 \times 0e \;\oplus\; 8 \times 1o \;\oplus\; 4 \times 2e, \tag{11}$$

with total component dimension $16 + 3(8) + 5(4) = 60$.

6 ACE-style body-ordered correlations

The density $\boldsymbol{A}_i$ is a two-body center–neighbor representation. Learned recursive Clebsch–Gordan products construct correlations up to configurable maximum body order $\nu_{\max}$:

$$\boldsymbol{C}_i^{(2)} = \boldsymbol{A}_i, \tag{12}$$

$$\boldsymbol{C}_i^{(\nu+1)} = \mathrm{TP}_{\nu+1}(\boldsymbol{C}_i^{(\nu)}, \boldsymbol{A}_i), \qquad 2 \le \nu < \nu_{\max}. \tag{13}$$

Each tensor product has independent learned multiplicity weights. This replaces the architecture-v1 all-ones contraction, which made nominal copies of the same irrep rank deficient. Because each input density is already a sum over the neighbor set, Eq. (13) is invariant to neighbor ordering and implicitly contains all repeated-index and distinct-neighbor terms admitted by the channel and angular truncations.

The node irreps are

$$64 \times 0e \ \oplus\ 32 \times 1o \ \oplus\ 16 \times 2e, \tag{14}$$

with component dimension 240. The initial center state is

$$\boldsymbol{h}_i^{(0)} = \mathrm{Center}(\boldsymbol{e}_i) + \sum_{\nu=2}^{\nu_{\max}} \mathrm{Linear}_\nu(\boldsymbol{C}_i^{(\nu)}). \tag{15}$$

Only scalar slots receive the explicit center embedding. Thus Z_i remains identifiable even if two centers have identical neighbor densities.

The construction is a compressed, learned ACE density-correlation basis. It is not claimed to be a formally complete, untruncated linear ACE basis. The ACEsuit/EquivariantTensors.jl construction enumerates sparse many-body (n, ℓ, m) specifications, applies symmetric products, and then uses sparse permutation-invariant Clebsch–Gordan symmetrization maps [5]. Transformers-ACE keeps the same $O(3)$ tensor-product rules and neighbor permutation invariance, but uses a learned low-rank density polynomial for scalability. Increasing $\ell_{\max}$, radial resolution, multiplicities, and correlation order expands the representation systematically but changes computational cost.

7 Strictly local equivariant attention

The attention block is local to one center and its fixed edge features. It does not read $\boldsymbol{h}_j^{(t)}$ from a sender atom. Scalar queries are generated from the pre-normalized center state, while keys use only scalar components of the fixed edge density:

$$\boldsymbol{q}_{ip} = W_p^Q \,\mathrm{LN}(\boldsymbol{h}_{i,\ell=0}^{(0)}), \tag{16}$$

$$\boldsymbol{k}_{ijp} = W_p^K \,\mathrm{LN}(\boldsymbol{a}_{ij,\ell=0}). \tag{17}$$

For head p, the invariant logit is

$$s_{ijp} = \frac{\boldsymbol{q}_{ip}^{\top}\boldsymbol{k}_{ijp}}{\sqrt{d_k}} + b_p(\boldsymbol{B}(d_{ij})) - \mathrm{softplus}(\lambda_p) d_{ij}, \qquad \widetilde{s}_{ijp} = s_{ijp}/T. \tag{18}$$

All terms are rotational invariants. The distance term is learned independently for each head and T is the training-time attention temperature.

7.1 Cutoff-null normalization

Ordinary softmax can cancel a cutoff when the final neighbor approaches r_{c}, because numerator and denominator vanish together. V2 adds a unit null/self channel:

$$\alpha_{ijp} = \frac{f_{\mathrm{c}}(d_{ij}) \exp(\widetilde{s}_{ijp})}{1 + \sum_{k \in \mathcal{N}(i)} f_{\mathrm{c}}(d_{ik}) \exp(\widetilde{s}_{ikp})}. \tag{19}$$

The code evaluates Eq. (19) with a per-center maximum subtraction for numerical stability. The null channel guarantees $\alpha_{ijp} \to 0$ when an edge exits the neighbor list.

Equivariant values are fixed-edge linear maps,

$$\boldsymbol{v}_{ijp} = \mathrm{Linear}_p^V(\boldsymbol{a}_{ij}), \tag{20}$$

and the residual update is

$$\boldsymbol{u}_i = \mathrm{Linear}^O\left[\frac{1}{H}\sum_{p=1}^{H}\sum_{j\in\mathcal{N}(i)} \alpha_{ijp}\boldsymbol{v}_{ijp}\right], \qquad \widetilde{\boldsymbol{h}}_i = \boldsymbol{h}_i^{(0)} + \boldsymbol{\gamma}_{\mathrm{attn}} \odot \boldsymbol{u}_i. \tag{21}$$

Both α_{ijp} and $\boldsymbol{a}_{ij}$ contain the cutoff, so the attention message vanishes at least quadratically in the envelope near r_{c}.

7.2 Angular feedback into scalar energy

An equivariant linear map cannot directly map $\ell > 0$ values to a scalar. V2 therefore forms atom-local invariant squared norms after the attention residual:

$$n_{ic\ell} = \sum_{m=-\ell}^{\ell} \left|\widetilde{h}_{icm}^{(\ell)}\right|^2, \qquad \ell > 0. \tag{22}$$

These invariants are concatenated with the scalar channels and passed through a scalar feed-forward network:

$$\boldsymbol{h}_{i,0} = \widetilde{\boldsymbol{h}}_{i,0} + \boldsymbol{\gamma}_{\mathrm{ffn}} \odot \mathrm{MLP}_{\mathrm{ffn}}\Big(\mathrm{LN}[\widetilde{\boldsymbol{h}}_{i,0}, \boldsymbol{n}_i]\Big). \tag{23}$$

Equation (22) lets interference among directional neighbor values affect energy while preserving $O(3)$ invariance. The default $\mathrm{CsPbI_3}$ model uses one attention block, two heads, layer-scale initialization 10^{-2}, and dropout probability 0.03.

8 Locality, equivariance, and scaling

For each center i, Eqs. (9)–(23) depend only on Z_i and the raw species and geometry of edges in $\mathcal{N}(i)$. A change to an updated hidden state $\boldsymbol{h}_j$ cannot alter center i. Thus stacking the implemented attention blocks does not increase the receptive field beyond r_{c}. The model is strictly local in the same operational sense as a finite-cutoff many-body potential.

Spherical harmonics, equivariant linear maps, Clebsch–Gordan products, scalar attention weights, and invariant norms preserve the transformation law of every channel. The atomic energy is therefore invariant to global rotations, reflections, translations, and atom indexing, while its gradient transforms as a vector. Runtime and memory scale as

$$\mathcal{O}(N\overline{n}\,C_{\mathrm{edge}}) + \mathcal{O}(NC_{\mathrm{CG}}), \tag{24}$$

where $\overline{n}$ is the average neighbor count. There is no iterative center-to-center message passing.

9 Energy, forces, and periodic stress

Only scalar channels enter the atomic-energy readout,

$$E_i = \mathrm{MLP}_E(\boldsymbol{h}_{i,0}), \qquad E = \sum_i E_i + E_{\mathrm{ref}}(\{Z_i\}). \tag{25}$$

The readout widths are $64 \to 64 \to 1$. The reference energy is either a fitted per-species linear baseline or a constant mean energy per atom; it is independent of positions and does not change forces or stress.

Forces are exact negative energy gradients,

$$\boldsymbol{F}_i = -\frac{\partial E}{\partial \boldsymbol{r}_i}. \tag{26}$$

No direct force head is used. Consequently the force field is conservative up to floating-point and neighbor-list tolerances.

Periodic stress is obtained by differentiating with respect to six symmetric strain parameters. Define

$$\boldsymbol{\epsilon}(\boldsymbol{\eta}) = \begin{pmatrix} \eta_1 & \eta_4 & \eta_5 \\ \eta_4 & \eta_2 & \eta_6 \\ \eta_5 & \eta_6 & \eta_3 \end{pmatrix}, \qquad \boldsymbol{F}_\epsilon = \boldsymbol{I} + \boldsymbol{\epsilon}, \tag{27}$$

and deform both positions and cell as row vectors,

$$\boldsymbol{r}_i' = \boldsymbol{r}_i \boldsymbol{F}_\epsilon, \qquad \boldsymbol{h}' = \boldsymbol{h}\boldsymbol{F}_\epsilon. \tag{28}$$

At zero strain, normal and shear components are reconstructed as

$$\sigma_{aa} = \frac{1}{V}\frac{\partial E}{\partial \eta_a}, \qquad a = 1, 2, 3, \tag{29}$$

$$\sigma_{xy} = \frac{1}{2V}\frac{\partial E}{\partial \eta_4}, \quad \sigma_{xz} = \frac{1}{2V}\frac{\partial E}{\partial \eta_5}, \quad \sigma_{yz} = \frac{1}{2V}\frac{\partial E}{\partial \eta_6}. \tag{30}$$

The factor $1/2$ appears because one shear parameter changes two symmetric matrix entries. $V = |\det \boldsymbol{h}'|$ is the deformed volume. This sign convention matches ASE's $\sigma_{ab} = V^{-1}\partial E/\partial \epsilon_{ab}$ convention; ASE cell filters supply the sign required for cell forces. The v2 architecture retains the validated v1 stress target conversion, Voigt ordering (xx, yy, zz, yz, xz, xy), and strain-gradient implementation unchanged.

10 Training objective

For structure s with N_s atoms, the supervised losses are

$$\mathcal{L}_E = \frac{1}{|\mathcal{B}|}\sum_{s\in\mathcal{B}}\left[\frac{E_s - E_s^{\mathrm{DFT}}}{N_s}\right]^2, \tag{31}$$

$$\mathcal{L}_F = \frac{1}{\sum_s 3N_s}\sum_{s,i,a}\left(F_{sia} - F_{sia}^{\mathrm{DFT}}\right)^2, \tag{32}$$

$$\mathcal{L}_\sigma = \frac{1}{6|\mathcal{B}_\sigma|}\sum_{s\in\mathcal{B}_\sigma}\sum_{v=1}^{6}\left(\sigma_{sv} - \sigma_{sv}^{\mathrm{DFT}}\right)^2, \tag{33}$$

$$\mathcal{L} = w_E\mathcal{L}_E + w_F\mathcal{L}_F + w_\sigma(t)\mathcal{L}_\sigma + w_{\mathrm{Sob}}\mathcal{L}_{\mathrm{Sob}}. \tag{34}$$

The light local linearization regularizer is

$$\mathcal{L}_{\mathrm{Sob}} = \left[E(\boldsymbol{r}+\boldsymbol{\delta}) - E(\boldsymbol{r}) + \sum_i \boldsymbol{F}_i \cdot \boldsymbol{\delta}_i\right]^2, \qquad \delta_{ia} \sim \mathcal{N}(0, \sigma_\delta^2), \tag{35}$$

with detached forces in this auxiliary term. The default configuration uses $w_E = 1$, $w_F = 10$, $w_\sigma : 1000 \to 100000$ linearly over the first 20 epochs, $w_{\mathrm{Sob}} = 10^{-3}$, and $\sigma_\delta = 0.02$ Å. The stress ramp avoids introducing large second-derivative optimization terms at random initialization.

Attention temperature decreases from 1.3 to 1.0 over 15 epochs. An optional force-error adaptation multiplies this schedule by $(\widehat{\mathcal{L}}_F^{1/2}/F_{\mathrm{ref}})^\gamma$, with $F_{\mathrm{ref}} = 0.5$ eV/Å and $\gamma = -0.25$ in the supplied $CsPbI_3$ configuration.

11 Data partitioning and optimization

Ordered molecular-dynamics frames are strongly correlated. A frame-random split can place adjacent configurations in training and validation, producing a misleading generalization estimate. The default split selects complete contiguous blocks of 25 frames for validation and removes a two-frame boundary gap from training. Model initialization, block selection, and data-loader shuffling use seed 42. An independent phase- and trajectory-grouped validation file remains preferable when metadata are available.

The default optimizer is AMSGrad Adam with learning rate 10^{-3}, weight decay 10^{-5}, gradient-norm clipping at 1, and cosine decay over 100 epochs to 5×10^{-5}. Batch size is eight. Validation checkpointing begins after the stress ramp; `model.pt` stores the best validation loss and `model_last.pt` stores the final state. Early stopping uses patience 20, minimum epoch 25, and minimum improvement 10^{-4}.

12 Default $CsPbI_3$ architecture

Quantity	Value
Cutoff / radial basis	6.0 Å / 12 Bessel functions
Maximum angular degree	$\ell_{\max} = 2$
Node irreps	$64 \times 0e + 32 \times 1o + 16 \times 2e$
Correlation irreps	$16 \times 0e + 8 \times 1o + 4 \times 2e$
Maximum correlation/body order	default 4; allowed 2–6
Radial network	$12 \rightarrow 32 \rightarrow$ TP weights
Attention	1 local block, 2 heads
Scalar FFN	invariant scalars and $\ell > 0$ squared norms
Readout	$64 \rightarrow 64 \rightarrow 1$
Trainable parameters	132,005
Direct force/stress heads	none
Long-range electrostatics	not included

Table 1: Architecture-version-2 defaults used by the supplied $CsPbI_3$ training configuration.

Relative to v1, the default v2 model reduces the parameter count from 593,097 to 132,005 by replacing the oversized radial/contraction parameterization with compact independent channels and one local attention block.

13 Verification and scientific use

Automated regression tests cover:

- sensitivity of a center representation to its own species;
- non-collapsed Clebsch–Gordan multiplicity channels;
- internal $O(3)$ equivariance and inversion parity of ACE tensor channels;
- value, first derivative, and second derivative of the cutoff;
- energy and force continuity when an edge leaves the neighbor list;
- rotation, translation, and atom-permutation behavior;
- absence of updated sender-state dependence in attention;
- finite-difference agreement for all symmetric stress components;
- deterministic blocked validation splitting; and
- strict loading of both legacy-v1 and versioned-v2 checkpoints.

For scientific deployment, low test RMSE is necessary but insufficient. A checkpoint should additionally reproduce phase-resolved errors, energy–volume curves, elastic constants, phonons, defect geometries, and stable NVE/NPT dynamics over the intended temperature and pressure range. Variable-cell relaxations are meaningful only after the stress sign, units, and finite-strain response have been validated against the reference electronic-structure method.

14 Limitations

The model is finite-cutoff and contains no explicit Coulomb, Ewald, charge equilibration, dispersion-tail, or reciprocal-space term. This is intentional for the present implementation. For ionic $CsPbI_3$, transferability with respect to cutoff, cell size, dielectric environment, charged defects, and long-wavelength distortions must therefore be tested explicitly. Long-range physics should be added only when controlled cutoff and equation-of-state tests demonstrate a deficiency that cannot be resolved by local data coverage.

The learned compressed correlations are not a proof of ACE basis completeness. The current validation split reduces temporal leakage but cannot replace truly independent trajectories and phases. Finally, architecture-v1 and v2 checkpoints are not weight-compatible. Checkpoints without an `architecture_version` field are evaluated by the legacy model; new training writes `architecture_version: 2` and cannot resume v1 weights.

Code availability and reproducibility

The implementation, training configuration, tests, and $CsPbI_3$ evaluation workflow are maintained at `https://github.com/paramvir3/Transformers-ACE`. The architecture equations in this document correspond to the version-2 classes `ACEV2Descriptor`, `StrictLocalEquivariantAttentionBlock`, and `TransformersACE`. Exact package versions and checkpoint hashes should be recorded for every reported result.

References

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